

QLDPC CODE, SURFACE CODE AND DECODING

{XIAONAN CHEN, GORDON LI, RAJDEEP TAH, TIANYUN TANG, JIA WANG} $\subseteq$ qLDPC-SURFACE CODE $\subseteq$ QEC

1. QUANTUM LDPC

1.1. Classical linear codes. A binary linear code of block length n is a subspace $C \subseteq \mathbb{F}_2^n$. If $\dim C = k$, then the code stores k information bits in n transmitted bits. Its minimum distance is

$$d(C) = \min_{0 \neq c \in C} \text{wt}(c), \quad (1.1)$$

where $\text{wt}(c)$ is the Hamming weight. We say that C has parameters $[n, k, d]$. Equivalently, a linear code may be specified by a parity-check matrix:

$$H \in \mathbb{F}_2^{m \times n}, \quad C = \ker H.$$

If a codeword $c \in C$ is corrupted by an error $e \in \mathbb{F}_2^n$, the receiver observes $y = c + e$. The syndrome is

$$Hy = H(c + e) = He, \quad (1.2)$$

so the unknown codeword disappears and the syndrome records information about the error.

The two basic asymptotic parameters are the rate and relative distance:

$$R = \frac{k}{n}, \quad \delta = \frac{d}{n}.$$

A family of codes is *asymptotically good* if R and δ are both bounded away from zero as $n \rightarrow \infty$. A family is *low-density parity-check* (LDPC) if the row and column weights of its parity-check matrices remain bounded independently of n . Thus each parity check involves only $O(1)$ symbols, and each symbol participates in only $O(1)$ checks.

1.2. Tanner codes and expander graph. Tanner codes [13] are a fundamental way to build classical LDPC codes from a constant-size local code and a sparse graph. Let

$$\Gamma = (L \sqcup R, E)$$

be a D -regular bipartite graph with $|L| = |R| = m$, $|E| = Dm$. Place one code symbol on every edge. Let $C_0 \subseteq \mathbb{F}_2^D$ be a fixed local code with parameters $[D, k_0, d_0]$. For $x \in \mathbb{F}_2^E$ and $v \in L \sqcup R$, write

$$x|_{E(v)} \in \mathbb{F}_2^D$$

for the restriction of x to the D edges incident with v . The Tanner code is

$$T(\Gamma, C_0) = \{x \in \mathbb{F}_2^E : x|_{E(v)} \in C_0 \text{ for every } v \in L \sqcup R\}. \quad (1.3)$$

If $C_0 = \ker h_0$, then the global parity-check matrix is obtained by placing one copy of h_0 at every vertex of Γ . Since D and h_0 are fixed, the resulting family is LDPC.

The rate can be estimated by counting constraints. There are Dm edge symbols, and each of the $2m$ vertices imposes at most $D - k_0$ independent constraints. Hence the global dimension K satisfies

$$K \geq Dm - 2m(D - k_0) = m(2k_0 - D), \quad (1.4)$$

and therefore any fixed local code of rate greater than $1/2$ gives a positive global rate.

The minimum distance is where expander graph enters. Let $x = \mathbf{1}_F \neq 0$ be a Tanner-codeword, and let

$$S = \{u \in L : E(u) \cap F \neq \emptyset\}, \quad T = \{v \in R : E(v) \cap F \neq \emptyset\}.$$

Since every active local view is a nonzero word of C_0 ,

$$|F| \geq d_0|S|, |F| \geq d_0|T| \implies |F| \geq d_0\sqrt{|S||T|}.$$

On the other hand, by expander mixing,

$$\begin{aligned} |F| &\leq e(S, T) \leq \frac{D}{m}|S||T| + \lambda\sqrt{|S||T|} \\ d_0\sqrt{|S||T|} &\leq \frac{D}{m}|S||T| + \lambda\sqrt{|S||T|} \\ d_0 - \lambda &\leq \frac{D}{m}\sqrt{|S||T|} \\ \sqrt{|S||T|} &\geq \frac{d_0 - \lambda}{D}m \\ |F| &\geq \frac{d_0(d_0 - \lambda)}{D}m. \end{aligned}$$

Since $N = |E| = Dm$, $d_0 > \lambda$ implies linear global distance. Therefore, whenever $d_0/D > \lambda/D$, the Tanner code has linear minimum distance. Conceptually,

local distance + global graph expansion $\implies$ linear global distance.

1.3. Quantum codes and some examples. The state space of n qubits is

$$\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}.$$

Definition 1.1. An $[[n, k, d]]$ quantum code is a 2^k -dimensional subspace $Q \subseteq \mathcal{H}_n$ that detects every error supported on fewer than d qubits.

Since Pauli operators form a basis of the operator space, it suffices to study Pauli errors. For a single qubit the Pauli operators are $\{I, X, Z, Y = iXZ\}$. Ignoring an overall phase, every n -qubit Pauli operator can be written as

$$X^a Z^b, \quad a, b \in \mathbb{F}_2^n. \quad (1.5)$$

Thus Pauli errors are represented by two binary vectors, one recording the X -part and one the Z -part. A stabilizer code is the simultaneous $+1$ -eigenspace of a commuting family of Pauli operators.

1.3.1. CSS codes. CSS codes [1, 12] are the special class in which the generators split into purely X -type and purely Z -type operators. Let

$$H_X \in \mathbb{F}_2^{m_X \times n}, \quad H_Z \in \mathbb{F}_2^{m_Z \times n}.$$

For every row x of H_X use the stabilizer X^x , and for every row z of H_Z use Z^z . Since

$$X^x Z^z = (-1)^{x \cdot z} Z^z X^x, \quad (1.6)$$

all stabilizers commute if and only if

$$H_X H_Z^T = 0. \quad (1.7)$$

The CSS code space is

$$Q = \{|\psi\rangle : X^x|\psi\rangle = |\psi\rangle \ \forall x \in \text{row}(H_X), \ Z^z|\psi\rangle = |\psi\rangle \ \forall z \in \text{row}(H_Z)\}. \quad (1.8)$$

Its number of encoded logical qubits is

$$k = n - \text{rank } H_X - \text{rank } H_Z. \quad (1.9)$$

Unlike classical distance, quantum distance is measured modulo stabilizers. An undetected Z -type operator satisfies $H_X z = 0$, but it is harmless if $z \in \text{row}(H_Z)$. Hence

$$d_Z = \min\{\text{wt}(z) : z \in \ker H_X \setminus \text{row}(H_Z)\}, \quad (1.10)$$

$$d_X = \min\{\text{wt}(x) : x \in \ker H_Z \setminus \text{row}(H_X)\}. \quad (1.11)$$

The distance is $d = \min\{d_X, d_Z\}$. A CSS family is qLDPC if both H_X and H_Z have bounded row and column weights.

The Calderbank–Shor–Steane framework thus reduces an important part of the quantum coding problem to a binary construction problem: combine two sparse classical constraint systems while enforcing the additional orthogonality condition (1.7) and retaining large rate and distance.

1.3.2. Hypergraph-product codes. The hypergraph product of Tillich and Zémor [14] gives a canonical way to enforce the CSS commutation constraint. Let

$$A \in \mathbb{F}_2^{m_A \times n_A}, \quad B \in \mathbb{F}_2^{m_B \times n_B}$$

be classical parity-check matrices. Define

$$H_X = (A \otimes I_{n_B} \quad I_{m_A} \otimes B^T), \quad H_Z = (I_{n_A} \otimes B \quad A^T \otimes I_{m_B}). \quad (1.12)$$

Then over $\mathbb{F}_2$,

$$H_X H_Z^T = A \otimes B^T + A \otimes B^T = 0.$$

If A and B are LDPC, then the resulting CSS code is also qLDPC.

The number of physical qubits is

$$N = n_A n_B + m_A m_B. \quad (1.13)$$

For suitable good classical LDPC constituents of comparable size L , the construction has positive quantum rate and minimum distance of order L . Since $N = \Theta(L^2)$, this yields

$$d = \Theta(L) = \Theta(\sqrt{N}). \quad (1.14)$$

Thus the hypergraph product solves the commutation and sparsity problems, but its natural product geometry leads to a square-root-distance regime.

1.4. The Panteleev–Kalachev breakthrough. Panteleev and Kalachev broke the square-root barrier by combining a product construction with a large nonabelian group symmetry.

Theorem 1.2 (Panteleev–Kalachev, 2021 [11]). For every fixed $R_0 \in (0, 1)$ and every finite field $\mathbb{F}_q$, there exists an explicit family of quantum LDPC codes with parameters $[[n, k, d]]_q$ such that

$$k \geq R_0 n, \quad d = \Theta(n).$$

In particular, there exist explicit asymptotically good binary qLDPC codes.

The algebraic mechanism is the *lifted product*. Suppose two classical Tanner-type codes admit a free action of a finite group G on their coordinates and checks. Let $\mathcal{R} = \mathbb{F}_q[G]$ be the group algebra. Their parity-check maps can then be written compactly as constant-size matrices

$$A \in \mathcal{R}^{m_a \times n_a}, \quad B \in \mathcal{R}^{m_b \times n_b},$$

where m_a, n_a, m_b, n_b remain constant while $|G|$ grows.

The lifted product is the tensor product over the common algebra $\mathcal{R}$. At the level of the associated three-term sequence, one has

$$\mathcal{R}^{n_a n_b} \xrightarrow{\partial_2} \mathcal{R}^{n_a m_b} \oplus \mathcal{R}^{m_a n_b} \xrightarrow{\partial_1} \mathcal{R}^{m_a m_b}, \quad (1.15)$$

with, schematically,

$$\partial_2 = \begin{pmatrix} A \otimes_{\mathcal{R}} I \\ -I \otimes_{\mathcal{R}} B \end{pmatrix}, \quad \partial_1 = (I \otimes_{\mathcal{R}} B \quad A \otimes_{\mathcal{R}} I).$$

The product identity $\partial_1 \partial_2 = 0$ is the analogue of $H_X H_Z^\top = 0$. Consequently the middle space of (1.15) contains only a constant number of copies of $\mathcal{R}$, and therefore

$$N = |G|(n_a m_b + m_a n_b) = \Theta(|G|). \quad (1.16)$$

An ordinary tensor product over $\mathbb{F}_q$ would retain two independent group coordinates and have size $\Theta(|G|^2)$.

The linear block length is not sufficient by itself: the difficult part is to show that nontrivial logical operators also have linear weight. The construction uses Tanner codes on highly symmetric Cayley expanders, together with a stronger local property called *product expansion*. Roughly, a local $w \times w$ pattern obtained as a sum of codeword-columns and codeword-rows cannot become very sparse merely because the row and column contributions cancel. In a suitable reduced representation, product expansion has the schematic form

$$\text{wt}(M) \gtrsim w (\#\{\text{active codeword-columns}\} + \#\{\text{active codeword-rows}\}). \quad (1.17)$$

This plays the role of local minimum distance in the classical Tanner-code argument. The global Cayley expansion then forces a locally nontrivial pattern to spread across a linear fraction of the code:

$\text{local product expansion} + \text{global Cayley expansion} \implies d = \Theta(n).$

1.5. Recent developments. The 2021 result settled the structural qLDPC conjecture, but it also changed the focus of the field. Once constant rate, linear distance, and bounded-weight checks were known to coexist, subsequent work began to ask for cleaner constructions and increasingly algorithmic properties:

- Leverrier and Zémor [9] introduced *quantum Tanner codes* in 2022. Their construction uses a left-right Cayley complex and makes the analogy with classical Tanner codes considerably more direct.
- The same year, Dinur, Hsieh, Lin, and Vidick [4] constructed explicit good qLDPC codes with linear-time decoders. Their result shows that good asymptotic parameters and sparse checks can coexist with an efficient classical syndrome-decoding algorithm capable of correcting a linear fraction of adversarial errors.
- A complementary expander route was proposed by Lin and Hsieh [10]: appropriate two-sided lossless expanders with a free group action would imply good qLDPC codes with linear-time decoding.
- Hsieh, Lubotzky, Mohanty, Reiner, and Zhang [7] resolved this graph-theoretic bottleneck in 2025 by constructing explicit constant-degree lossless vertex expanders with the required symmetry. This did not establish the first explicit good qLDPC code; rather, it completed a particularly clean lossless-expander route to such codes.
- More recently, Wills, Lin, Zhang, and Hsieh [16] studied a different complexity measure. Their 2026 construction gives asymptotically good quantum codes with linear total gate complexity for encoding and unencoding and logarithmic encoding depth, together with fast classical decoding. These codes are not asserted to be qLDPC and are analyzed in a channel-coding setting in which the encoding and unencoding operations themselves are ideal. The result is therefore complementary to the qLDPC line.

The present landscape is thus no longer an existence problem alone. Good qLDPC codes and linear-time syndrome decoders are known, while active questions ask how locality, encoding complexity, decoding, fault tolerance, logical operations, and stronger notions such as local testability can be optimized simultaneously.

2. SURFACE CODE AND DECODING

2.1. The planar surface code. The surface code is a two-dimensional CSS stabilizer code and one of the most important examples of a geometrically local quantum low-density parity-check (qLDPC) code. In the CSS formulation, the code is described by two families of commuting parity checks: X-type checks and Z-type checks. In the surface code, these abstract checks are realized as local operators on a two-dimensional lattice [2].

For the unrotated planar surface code, physical data qubits are placed on the edges of a square lattice. Ancilla qubits are used to measure two types of local stabilizer generators. The X-type stabilizer associated with a vertex v is,

$$A_v = \prod_{e \ni v} X_e, \quad (2.1)$$

where the product is taken over the data-qubit edges incident on v . Similarly, the Z-type stabilizer associated with a plaquette p is

$$B_p = \prod_{e \in \partial p} Z_e, \quad (2.2)$$

where ∂p denotes the set of data-qubit edges forming the boundary of p . Stabilizers near a physical boundary generally act on fewer data qubits than stabilizers in the bulk [2].

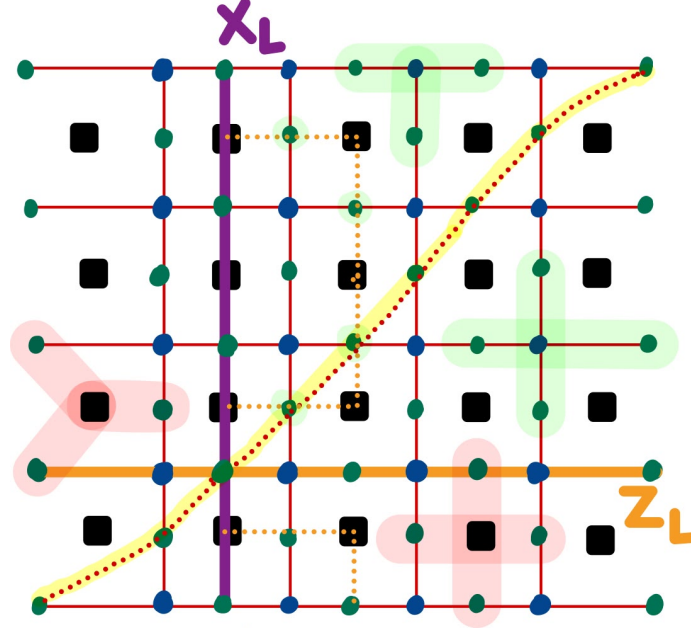


FIGURE 1. Unrotated planar surface code. *Green* circles represent physical data qubits, *blue* ancillas measure X-type stabilizers, and *black* ancillas measure Z-type stabilizers. The *purple* and *orange* strings represent examples of the logical operators $\bar{X}$ and $\bar{Z}$, respectively.

The code space, denoted by $\mathcal{C}$, is the simultaneous $+1$ eigenspace of all stabilizer generators:

$$\mathcal{C} = \{|\psi_L\rangle : A_v|\psi_L\rangle = |\psi_L\rangle, B_p|\psi_L\rangle = |\psi_L\rangle \text{ for all } v, p\}. \quad (2.3)$$

The stabilizer measurements reveal local error parities without directly measuring the encoded logical state. In the absence of errors, every stabilizer measurement has eigenvalue $+1$. A physical Pauli error that anticommutes with a stabilizer changes its measured eigenvalue from $+1$ to -1 , thereby producing syndrome information [2].

The surface code satisfies the LDPC property because every stabilizer acts on only a constant number of neighboring data qubits, and every data qubit participates in only a constant number of stabilizer checks, independently of the lattice size. However, a single planar surface-code patch is not an asymptotically good qLDPC family: it encodes one logical qubit and has distance proportional to the linear size of the lattice, so that $d = \Theta(\sqrt{n})$ [2].

Logical information is stored *nonlocally*. Representative logical Pauli operators may be written as string operators,

$$\bar{X} = \prod_{e \in \gamma_X} X_e, \quad \bar{Z} = \prod_{e \in \gamma_Z} Z_e, \quad (2.4)$$

where γ_X and γ_Z connect the appropriate opposite boundaries of the planar patch. These operators commute with every stabilizer but cannot themselves be expressed as products of stabilizers [2].

Multiplication by stabilizers can continuously deform the path of a logical string without changing its action on the encoded state. However, a nontrivial boundary-to-boundary logical string cannot be removed completely or contracted to a local operator. If γ_X and γ_Z intersect on an odd number of data qubits, then the corresponding logical operators anticommute:

$$\bar{X}\bar{Z} = -\bar{Z}\bar{X}. \quad (2.5)$$

They therefore act as the ordinary Pauli X and Z operators on the encoded logical qubit. For an unrotated planar surface code of distance d , the number of data qubits is [2]

$$n = d^2 + (d - 1)^2. \quad (2.6)$$

For the distance-five patch shown in Fig. 1,

$$n = 5^2 + 4^2 = 41. \quad (2.7)$$

The code therefore has parameters,

$$[[n, k, d]] = [[41, 1, 5]], \quad (2.8)$$

meaning that 41 physical data qubits encode one logical qubit with code distance five. A distance- d code can correct every physical Pauli error of weight at most,

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor. \quad (2.9)$$

Thus, the distance-five patch can correct every physical Pauli error acting on at most *two* data qubits.

2.2. Error chains and syndrome defects. Physical errors are detected through the stabilizer outcomes. For example, a Pauli-Z error anticommutes with each neighboring X-type stabilizer and therefore changes the corresponding stabilizer outcomes. Similarly, an X-type error is detected by neighboring Z-type stabilizers.

Consider a connected chain E of Z-errors. An X-type stabilizer adjacent to two errors in the interior of the chain is flipped twice. Since the syndrome is evaluated modulo two, these two contributions cancel. Consequently, only the checks located at the endpoints of the error chain remain violated. These violated checks are called *syndrome defects* [2].

In the geometric chain description, the measured syndrome may be identified with the boundary of the error chain:

$$s = \partial E. \quad (2.10)$$

An error chain may also terminate at an appropriate physical boundary of the planar code. In that case, only one endpoint appears as a bulk syndrome defect. This motivates the introduction of virtual boundary vertices in the decoding graph [2].

The syndrome identifies the endpoints of an error chain, but it does not uniquely determine the path followed by the physical errors. Different chains can have the same boundary and therefore produce the same syndrome:

$$\partial E_1 = \partial E_2 = s. \quad (2.11)$$

The decoding problem is therefore to infer an error chain that is both consistent with the observed syndrome and sufficiently probable under the assumed physical noise model. In the code-capacity noise model, only the data qubits are noisy and all stabilizer measurements are assumed perfect.

The geometric decoding problem can be represented by a weighted graph. Observed syndrome defects form graph vertices, while an edge represents a possible elementary error mechanism or a path of such mechanisms. For an independent error mechanism e occurring with probability p_e , one assigns the weight,

$$w_e = \log \left(\frac{1 - p_e}{p_e} \right). \quad (2.12)$$

For a candidate error chain E , the total weight is,

$$W(E) = \sum_{e \in E} w_e. \quad (2.13)$$

More probable error mechanisms have smaller weights. Minimum-weight perfect matching pairs each observed syndrome defect either with another defect or with an appropriate virtual boundary, and selects the pairing whose associated paths have minimum total weight [2, 6].

The decoder is not required to reconstruct the exact physical error. It must instead select a correction belonging to the same logical equivalence class as the actual error. If the actual error and the decoder correction differ only by stabilizers, the logical state is preserved. If their combination contains a nontrivial logical string, the syndrome may be removed even though a logical error has occurred [2, 8].

The following subsection reformulates this geometric picture algebraically using a detector error model, which specifies the detectors and logical observables flipped by each elementary error mechanism.

2.3. Detector error models and most-likely-error decoding. A detector error model (DEM) describes a noisy error-correction experiment in terms of independent error mechanisms and the detectors and logical observables that they flip [3, 5]. In matrix form, a DEM is specified by a triple

$$(H, p, L),$$

where $H \in \mathbb{F}_2^{m \times n}$ is the detector matrix, $p = (p_1, \dots, p_n)$ contains the probabilities of the n error mechanisms, and $L \in \mathbb{F}_2^{k \times n}$ is the logical map [6, 15]. An error configuration is represented by $x \in \mathbb{F}_2^n$, with $x_j = 1$ when error mechanism j occurs. Its detector outcome $s \in \mathbb{F}_2^m$ and logical effect $\ell \in \mathbb{F}_2^k$ are

$$s = Hx, \quad \ell = Lx, \quad (2.14)$$

with all arithmetic performed over $\mathbb{F}_2$. Assuming that the error mechanisms occur independently,

$$\mathbb{P}(x) = \prod_{j=1}^n (1 - p_j)^{1-x_j} p_j^{x_j}. \quad (2.15)$$

Given an observed detector outcome s , most-likely-error decoding selects a most probable individual error configuration consistent with that outcome [8, 15]:

$$\hat{x}^{\text{MLE}}(s) \in \underset{\substack{x \in \mathbb{F}_2^n \\ Hx=s}}{\text{argmax}} \mathbb{P}(x). \quad (2.16)$$

The inclusion symbol allows for the possibility of several equally likely maximizers. For independent error mechanisms, this optimization is equivalently written as

$$\hat{x}^{\text{MLE}}(s) \in \underset{\substack{x \in \mathbb{F}_2^n \\ Hx=s}}{\operatorname{argmin}} \sum_{j=1}^n w_j x_j, \quad w_j = \log\left(\frac{1-p_j}{p_j}\right). \quad (2.17)$$

For graphlike DEMs, this weighted problem gives the decoding problem addressed by minimum-weight perfect matching (MWPM) [2, 6]. Once an error estimate has been selected, the decoder assigns its corresponding logical effect,

$$\hat{\ell}^{\text{MLE}}(s) = L\hat{x}^{\text{MLE}}(s). \quad (2.18)$$

Example 2.1 (One-round surface-code DEM). Consider a single round of perfect X-stabilizer measurements in the presence of Z errors on the 13 data-qubit locations $q_1, \dots, q_{13}$ as in Fig 2. We identify the j th DEM error mechanism with a Z error on q_j , and write the error configuration as

$$x = (x_1, \dots, x_{13})^T \in \mathbb{F}_2^{13},$$

where $x_j = 1$ indicates that a Z error occurred on q_j .

Ordering the six X-stabilizer detectors as

$$(X_{\text{BL}}, X_{\text{BR}}, X_{\text{ML}}, X_{\text{MR}}, X_{\text{TL}}, X_{\text{TR}}),$$

the detector outcome $s \in \mathbb{F}_2^6$ is given by

$$s = Hx,$$

where the detector matrix $H \in \mathbb{F}_2^{6 \times 13}$ is

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

Thus, for example, a Z error on q_2 flips the two bottom detectors, whereas a Z error on q_7 flips the two middle detectors.

There is one encoded logical qubit, so the logical outcome is a single bit

$$\ell \in \mathbb{F}_2.$$

Choosing the rightmost boundary support $\{q_3, q_8, q_{13}\}$ as the representative defining the logical map gives

$$\ell = Lx,$$

with

$$L = (0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1) \in \mathbb{F}_2^{1 \times 13}.$$

Consequently,

$$\ell = x_3 + x_8 + x_{13} \pmod{2}.$$

We illustrate the decoding procedure using the syndrome Fig 3

$$s = (1, 1, 0, 0, 1, 0)^T,$$

for which the detectors $X_{\text{TL}}, X_{\text{BL}},$ and X_{BR} are triggered. Two configurations satisfying $Hx = s$ are

$$x^{(1)} = e_{11} + e_2, \quad x^{(2)} = e_9 + e_4 + e_3.$$

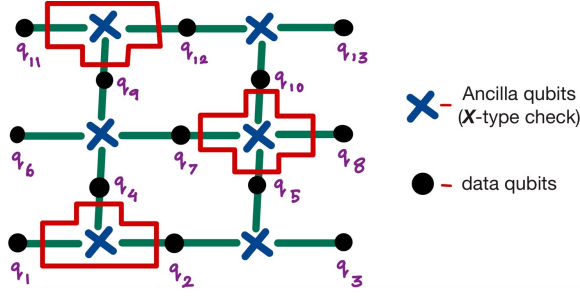


FIGURE 2. Schematic of a planar surface-code lattice. Black circles represent the data qubits $q_1, \dots, q_{13}$, blue crosses denote stabilizer checks, and green edges indicate nearest-neighbor couplings. Red contours highlight representative bulk and boundary stabilizer operators.

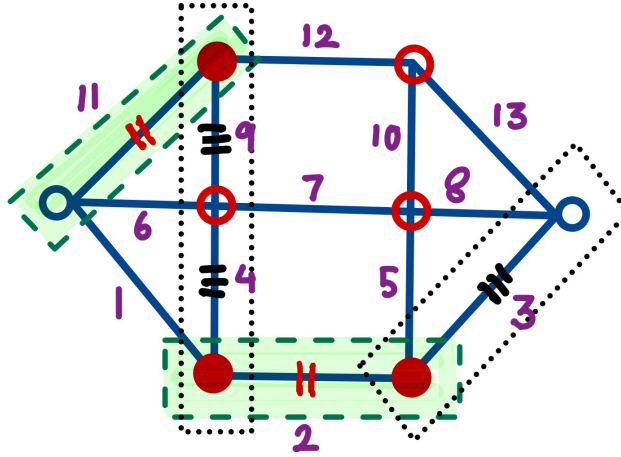


FIGURE 3. One-round surface-code DEM for the syndrome in which X_{TL} , X_{BL} , and X_{BR} are triggered. The configurations $e_{11} + e_2$ and $e_9 + e_4 + e_3$ produce the same syndrome, but the former has lower weight when all error mechanisms are equally weighted.

In the second configuration, the errors on q_9 and q_4 flip X_{ML} twice, so this detector does not appear in the syndrome. Assuming equal edge weights,

$$w(x^{(1)}) = 2w < 3w = w(x^{(2)}),$$

and hence the optimization in (2.17) gives

$$\hat{x}^{\text{MWPM}}(s) = e_{11} + e_2.$$

A logical ambiguity can arise when two configurations have the same syndrome and minimum weight. For example, the configurations shown in Fig. 4,

$$x^{(a)} = e_{12} + e_{10}, \quad x^{(b)} = e_{11} + e_8,$$

are equally weighted and satisfy $Hx^{(a)} = Hx^{(b)}$, but have different logical labels, $Lx^{(a)} = 0$ and $Lx^{(b)} = 1$. MWPM can select either $x^{(a)}$ or $x^{(b)}$ as in (2.18),

$$\hat{\ell}(s) = L\hat{x}(s).$$

In simulation, the sampled error x is known, and decoding succeeds when

$$L(x + \hat{x}(s)) = Lx + \hat{\ell}(s) = 0.$$

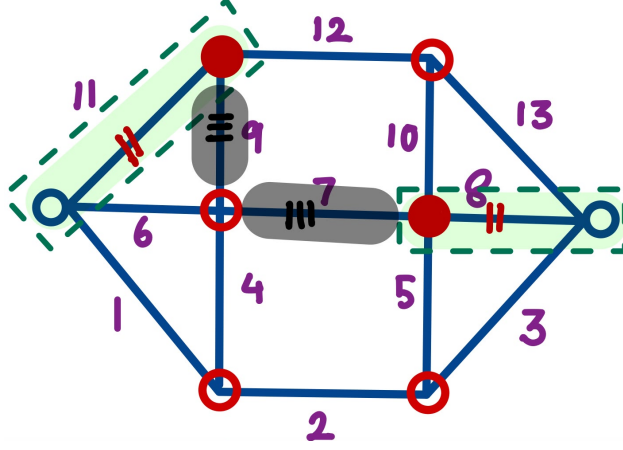


FIGURE 4. Logical ambiguity in MWPM: the equal-weight configurations $x^{(a)} = e_{12} + e_{10}$ and $x^{(b)} = e_{11} + e_8$ have the same syndrome but different logical labels.

In an experiment, x is not available to the decoder. A final logical measurement relative to a known prepared logical state can reveal whether a logical failure occurred, but this measurement is generally destructive and therefore serves to evaluate the decoder rather than to select a correction while preserving the encoded state.

REFERENCES

- [1] A. R. Calderbank and P. W. Shor. Good quantum error-correcting codes exist. *Physical Review A*, 54(2):1098–1105, 1996.
- [2] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill. Topological quantum memory. *Journal of Mathematical Physics*, 43(9):4452–4505, 2002.
- [3] P.-J. H. S. Derks, A. Townsend-Teague, A. G. Burchards, and J. Eisert. Designing fault-tolerant circuits using detector error models. *Quantum*, 9:1905, 2025.
- [4] I. Dinur, M.-H. Hsieh, T.-C. Lin, and T. Vidick. Good quantum ldpc codes with linear time decoders, 2022.
- [5] C. Gidney. Stim: a fast stabilizer circuit simulator. *Quantum*, 5:497, 2021.
- [6] O. Higgott and C. Gidney. Sparse blossom: Correcting a million errors per core second with minimum-weight matching. *Quantum*, 9:1600, 2025.
- [7] J.-T. Hsieh, A. Lubotzky, S. Mohanty, A. Reiner, and R. Y. Zhang. Explicit lossless vertex expanders, 2025.
- [8] P. Iyer and D. Poulin. Hardness of decoding quantum stabilizer codes. *IEEE Transactions on Information Theory*, 61(9):5209–5223, 2015.
- [9] A. Leverrier and G. Zémor. Quantum tanner codes, 2022.
- [10] T.-C. Lin and M.-H. Hsieh. Good quantum ldpc codes with linear time decoder from lossless expanders, 2022.
- [11] P. Panteleev and G. Kalachev. Asymptotically good quantum and locally testable classical ldpc codes, 2021.
- [12] A. M. Steane. Error correcting codes in quantum theory. *Physical Review Letters*, 77(5):793–797, 1996.
- [13] R. M. Tanner. A recursive approach to low complexity codes. *IEEE Transactions on Information Theory*, 27(5):533–547, 1981.
- [14] J.-P. Tillich and G. Zémor. Quantum ldpc codes with positive rate and minimum distance proportional to $n^{1/2}$, 2009.
- [15] Z. Tobias, N. P. Breuckmann, and B. Placke. Optimal decoding with the worm, 2026.
- [16] A. Wills, T.-C. Lin, R. Y. Zhang, and M.-H. Hsieh. Linear-time encodable and decodable quantum error-correcting codes, 2026.